

Fourteen Structural Barriers of ECH Bounce Cosmology

#	Barrier	Branch	Physical mechanism	Failure mode
1	Mass-coupling lock	A	m and g locked by single PGT parameter t_3	I
2	Topological-shift duality	B	Shift symmetry $\leftrightarrow$ geometric content exclusive	I
3	Scalar-tensor universality	C	$T_0 = Q_0 = 0$ on FRW; only $R(g)$ survives	I
4	Planck suppression	D	$1 \partial\phi/\text{vertex}$; disformal needs 2; $\text{dim-8} \sim 10^{-122}$	I
5	Scale separation	E	Bounce $\sim 10^{-32}$ of spacetime volume integrals	II
6	Attractor-sensitivity	F	Attractors erase ϕ ; otherwise 10^{-30} precision needed	II
7	Parameter immunity	G	Action params $\neq$ boundary conditions; cyclic incompatible	II
8	Parity-even interaction	H	$(J^5)^2$ is parity-even; $\overline{R}R = 0$ on FRW	III
9	Phase-space conservation	J	Liouville theorem at $H = 0$; no contraction	IV
10	UV-IR specificity	L	Generic mechanisms don't need the bounce	III
11	Decoupling universality	L	Light torsion: $g_{\text{eff}} \sim m_T / M^2_{\text{Pl}}$	I
12	Vacuum amplification ceiling	M	$\Omega_{\text{GW}} \propto (H_b/M_{\text{Pl}})^2$; gap $\geq 10^{17}$ to detectors	III
13	Gravitational democracy	N	Torsion $\sim 1\%$ of Planck-era channels; B , L conserved	V
14	Bounce-vacuum decoupling	O	Trigger (bounce) $\neq$ outcome (ρ_{DE})	V

Failure modes:

I: Geometric specificity lost on FRW Scale separation (10^{122} hierarchy) Generic or undetectable IV: Hamiltonian conservation V: Gravitational democracy